

SCI225: Pathophysiology

Week 1 Webinar — Full Speaker Script

Approximate run time: 60 minutes

Nightingale College

How to use this script

This is a complete word-for-word script you can read aloud while moving through the slide deck. It is paced for a full 60-minute webinar with a small buffer for student questions.

- Bold-italic [Stage cue] lines are directions for you — not for reading aloud.
- Each slide block lists a target time so you can pace yourself; cumulative timestamps are listed under the section banners.
- Replace the bracketed placeholders ([your name], [day/time], etc.) before the session.
- Built-in pauses are noted as “(pause)”. Resist the urge to skip them — they give learners time to absorb.

Section 1 • Welcome & Course Logistics (0:00 – 0:14)

Slide 1: SCI225: Pathophysiology — Week One (Title) (0:00 – 0:02 • 2 min)

[Stage cue] Start recording. Smile. Greet learners as they come in. Wait until the timestamp shows you are at the top of the hour, then begin.

Hello everyone, and welcome — welcome to SCI225, Pathophysiology. My name is Professor [your name], and I am so genuinely glad you are here for our very first session of the semester.

If you are joining a couple of minutes late, that is totally fine — we are just getting settled. Take a breath, get your notebook out, and let me know in the chat what part of the country, or the world, you are joining from. I love seeing where everyone is calling in from.

[Stage cue] Read out two or three locations from the chat to acknowledge people.

Beautiful. Alright — let's officially get started. This is the Week One webinar for Pathophysiology, and over the next hour we are going to do three things: meet each other, learn how this course works, and get our first real taste of pathophysiology — the science of how disease changes the body.

Slide 2: Agenda (0:02 – 0:03 • 1 min)

Here is our roadmap for tonight. We will start with a few housekeeping items — recordings, captioning, that sort of thing. Then I'll introduce myself, walk you through Canvas and where everything lives, talk about the assignments for Week 1, and then we will dive into the actual content: what is pathophysiology, a refresher on DNA and genes, mutations and inheritance, epigenetics, and finally cellular injury, fluids and electrolytes, and acid-base balance.

It is a lot — but I promise we will move at a pace that gives you time to think and to ask questions. Drop questions in the chat any time, and I will pause every few slides to address them.

Slide 3: Reminders Before We Start (0:03 – 0:05 • 2 min)

Three quick housekeeping items before we get into the content.

First — this webinar is being recorded. Anything you say on the microphone, anything you turn your camera on for, and anything you type into the chat may end up in the recording, which is posted for classmates who could not attend live. So please participate in whatever way feels comfortable to you. If you would rather keep your camera off and use the chat, that is completely fine. If you want to come on mic and ask a question out loud, that is wonderful too.

Second — captioning. If reading along helps you, you can turn on live captions at any time. Look at the top menu of your webinar window, click on “More,” and then click on “Captions.” You can toggle them on and off, and you can pull up the full transcript while we are talking.

Third — interaction. I want you to interact. You have several tools: the text chat, reactions like the raise-hand and thumbs-up, and your microphone and webcam. Please use them. I am going to ask you questions, and a quiet webinar is a long webinar.

[Stage cue] Pause and smile.

Alright — onto introductions.

Slide 4: Introduction — Your Instructor (0:05 – 0:07 • 2 min)

So a quick word about me. I am Professor [your name]. My background is in [briefly: clinical area, years teaching, why you love this subject]. The reason I teach pathophysiology is honestly because I think it is the most useful course you will take in nursing school — once you understand how the body breaks, everything else in your clinicals starts to click.

Here is how to reach me. My email is [your email]. The fastest way to get a response is through email — I generally answer within 24 hours, sometimes sooner, and I do my best not to leave you hanging over a weekend.

My phone number is on the slide if you ever need to reach me by phone, but email is preferred.

Office hours — I have posted a Calendly link, also on this slide. Please book a slot any time you want to talk through content, talk through an assignment, or just check in. Office hours are not a punishment. They are not just for people who are struggling. Some of my best conversations every semester are with students who come in just to talk through a tricky concept.

And then webinars — I run my live webinars on [day] at [time] Mountain Time. You are at one right now. There are also other faculty webinars in this course, and you are welcome to attend those if your schedule changes.

Slide 5: Important Dates and Times (0:07 – 0:08 • 1 min)

A really important slide. Nightingale College runs on Mountain Time. Mountain Time. I am going to repeat that because every semester someone misses an assignment because they were thinking in their local time zone.

Our school week runs from 12:00 a.m. Mountain Time on Monday through 11:59 p.m. Mountain Time on Sunday. So when an assignment is due “Sunday at 11:59 p.m.,” that is Mountain Time — not Eastern, not Pacific, Mountain.

If you live in another time zone, do yourself a favor right now: pull up the world clock on your phone or your laptop and add Mountain Time. It will save you many points and a lot of stress.

Our semester runs from [start date] through [end date], and the very last assignment closes at 5:00 p.m. Mountain Time on the final day.

Slide 6: Navigating the Canvas Classroom (0:08 – 0:10 • 2 min)

[Stage cue] Optional: share your screen and walk through Canvas live. If you do, briefly demonstrate: locating the home page, the modules, where to find webinar links, the textbook tile, and the office-hours booking link. If you are not screen-sharing, narrate the slide image instead.

Canvas is the home base for this course. Everything you need — the syllabus, the modules, the textbook, the assignments, the discussions, my office hours link — lives in Canvas.

When you log in and open SCI225, you will land on the course home page. From there, the most important buttons are along the left side: Modules, where the weekly content is laid out; Grades, where you can see your scores; the Webinar Links page, where you find the live sessions and recordings; and the Textbook tile, which takes you straight into the e-book.

If at any point this week you log in and you can't find something — please email me. Don't spend an hour searching. I would much rather get a quick email and point you to the right place than have you fall behind.

Slide 7: Canvas Priorities (0:10 – 0:11 • 1 min)

If you only do four things in Canvas each week, do these four.

- One — read the module introduction. It tells you what to focus on.
- Two — complete the readings before the webinar so you get the most out of our live time together.
- Three — attend a webinar. Live is best, but the recording counts too.
- Four — submit the case study and quiz before Sunday at 11:59 p.m. Mountain.

If you build a habit around those four steps every single week, you will succeed in this course.

Slide 8: How to Access the Webinars (0:11 – 0:12 • 1 min)

Quick walkthrough on how to access webinars, because this is the question I get asked the most in the first two weeks.

From the course home page, click HOME, then click Webinar Links. Click the SCI tab, then click SCI225: Pathophysiology. You will see a list of webinars with day and time. Click the link under “Webinar Link” to join the live session.

And here is the part everyone misses: if you scroll all the way down on that page, you will see a section for webinar recordings. So if you cannot make a live session, the recording will be there usually within 24 hours.

Slide 9: Learner Tools (0:12 – 0:13 • 1 min)

[Stage cue] Optional: briefly point out the tools available in the Learner Tools tab — academic coaching, tutoring, the writing center, library resources, accommodations.

Nightingale gives you a really robust set of support tools, and I want you to know they exist before you need them. Academic coaching, tutoring, the writing center, the library, and accommodations through the disability services office — they are all here, and they are all free.

If English is not your first language, if you have a learning difference, if you just want a second pair of eyes on a paper — please use these. They exist for you.

Slide 10: Weekly Resources (0:13 – 0:13 • 30 sec)

Each week inside the module, you will see a tile with weekly resources — slide handouts, supplemental videos, and study guides. Click on the module introduction first, every week, to get the lay of the land.

Slide 11: Weekly Study Guides (0:13 – 0:14 • 1 min)

This may be the single most valuable resource I can point you to. Each week we have a study guide — a structured set of questions and key concepts that map directly to the learning objectives.

If you can answer the study guide questions confidently, you are ready for the quiz. So a really effective workflow is: read the chapters, attend the webinar, fill out the study guide, then take the quiz. In that order.

[Stage cue] Optional: pause for chat questions before moving into Week 1 content. 30–60 seconds.

Section 2 • Week One Overview (0:14 – 0:19)

Slide 12: Week One (0:14 – 0:15 • 1 min)

Alright — that wraps up the logistics. Take a breath. We have officially started Week One of Pathophysiology.

Here is a fun fact before we dive in: you are about to spend an hour talking about why bodies break. By the end of tonight, you will be able to look at a person — or a chart — and start asking the right questions about what is going wrong at the cellular level. That is a real skill, and it is the foundation of every clinical decision you will ever make as a nurse.

Slide 13: Learning Objectives (0:15 – 0:16 • 1 min)

By the end of this week — and by the end of this webinar — you should be able to do four things.

- One — define pathophysiology and explain why it sits at the intersection of biology, suffering, and clinical care.
- Two — describe the structure and function of DNA, and identify the major categories of genetic and epigenetic disease.
- Three — explain altered cellular and tissue biology, including atrophy, hypertrophy, hyperplasia, dysplasia, metaplasia, and the difference between necrosis and apoptosis.
- Four — interpret basic alterations in fluid balance, electrolytes, and acid-base status.

That is a lot, but every single piece of that will come back, again and again, throughout your career. So this is the week to lock in the fundamentals.

Slide 14: What Is Due This Week? (0:16 – 0:19 • 3 min)

Let's talk about what is due this week. I am going to walk you through this carefully because the first week sets the rhythm for the rest of the semester.

- First — complete the Week 1 Preassessment. This is a low-stakes quiz that helps me see where everyone is starting from. It is not graded for correctness, but it is graded for completion.
- Second — read four chapters from your textbook: Chapter 2 on Genes and Genetic Diseases, Chapter 3 on Epigenetics and Disease, Chapter 4 on Altered and Injured Cells, and Chapter 5 on Fluids and Electrolytes, Acids and Bases. I know that is a lot for the first week. Pace yourself — read one chapter a night for four nights, and you will get through it.
- Third — attend a Week 1 webinar. You are doing that right now. Gold star.
- Fourth — complete the Week 1 case study. The case study is where you take what you have read and apply it to a patient scenario.

- And fifth — take the Week 1 webinar quiz. The quiz pulls directly from the material we are going through tonight, so if you take good notes during this hour, you are doing yourself a favor.

Everything is due Sunday at 11:59 p.m. Mountain Time. Build that into your week now — block off time on your calendar for the readings, time for the case study, and time for the quiz. Nursing school is a calendar problem more than it is a smart-or-not-smart problem.

Section 3 • What Is Pathophysiology + Genetics (0:19 – 0:38)

Slide 15: What Is Pathophysiology? (0:19 – 0:22 • 3 min)

Okay — content time. Let's start with the word itself. Pathophysiology is a Greek mash-up. Three roots.

- “Patho” comes from pathos, which means suffering.
- “Logos” means discourse, or the formal study of something.
- “Physio” means the functions of an organism — how things work.

So put those together: pathophysiology is the formal study of how an organism's functions change in the presence of suffering or disease. It is the study of what goes wrong in the body — at the molecular level, the cellular level, the tissue level, and the organ-system level — when a person becomes sick or injured.

Now I want you to notice something about this definition. The word “suffering” is right there in the root. Pathophysiology is not just chemistry and biology — it is chemistry and biology happening to a human being who is scared, in pain, has a family, has a job, has spiritual beliefs, and has a financial situation.

That last point matters. As a nurse, you cannot separate the molecular changes in a heart cell from the patient who feels their chest tightening and is terrified they are going to die before they can pay this month's rent. The science of pathophysiology and the human experience of disease are inseparable, and we are going to keep coming back to that idea all semester.

[Stage cue] Pause. Take a beat. Let it land.

Slide 16: Section Objectives — Genetics & Epigenetics (0:22 – 0:23 • 1 min)

Now we shift into our first content area — genetics and epigenetics. Two objectives for this section.

- One — understand the vast array of human genes and genetic diseases.
- Two — define epigenetics and recognize epigenetic-related disease.

To get there, we need to do a quick refresher on DNA, because everything that follows depends on it.

Slide 17: DNA (0:23 – 0:25 • 2 min)

Quick check-in. Without me telling you — what is DNA?

[Stage cue] Pause for chat responses (15–20 seconds). Read out a couple of answers.

Great answers. So DNA — deoxyribonucleic acid — is the instruction manual for building and running every cell in your body. It is a long, double-stranded molecule, twisted into a helix, and it lives inside the nucleus of almost every cell.

Components: DNA is made of nucleotides. Each nucleotide has three parts — a sugar, called deoxyribose; a phosphate group; and one of four nitrogenous bases. The bases are adenine, thymine, cytosine, and guanine — A, T, C, and G.

Base pairing: A always pairs with T, and C always pairs with G. That pairing is the magic. It is what allows DNA to be copied accurately, because if you separate the two strands, each strand is a perfect template for rebuilding its partner.

And what is DNA responsible for? It codes for proteins. Proteins do almost all the work in your body — they are enzymes, they are receptors, they are structural elements, they are signaling molecules. So when the DNA changes, the protein changes, and when the protein changes, the function changes. That cascade — DNA to protein to function — is the heart of genetic disease.

Slide 18: Replication, Transcription, Translation (0:25 – 0:27 • 2 min)

Three processes you should be able to define cold.

- Replication is making a copy of DNA. It happens before a cell divides. The double helix unzips, and each strand is used as a template to build a new complementary strand. You end up with two identical DNA molecules.
- Transcription is making messenger RNA — mRNA — from DNA. Inside the nucleus, a section of DNA is unzipped, and an enzyme called RNA polymerase reads one strand and builds an mRNA copy. That mRNA then leaves the nucleus.
- Translation is making protein from mRNA. The mRNA is read by a ribosome, three bases at a time. Each three-base codon corresponds to one amino acid. The ribosome strings amino acids together, and that chain folds into a functional protein.

So the central dogma — and this is a phrase you will hear forever — is DNA makes RNA makes protein. Replication, transcription, translation. Lock those words in. They are going to come back tonight when we talk about mutations.

Slide 19: Keeping Track of the Numbers (0:27 – 0:28 • 1 min)

A quick numbers-check, because students mix these up on quizzes.

- Humans have 46 chromosomes in our somatic cells — that is, in every regular body cell. We say we are diploid, which means our chromosomes come in 23 pairs.
- Our gametes — the sex cells, sperm and oocytes — have only 23 chromosomes. We call that haploid. They are haploid because of a special cell division called meiosis, which cuts the chromosome number in half.
- In males, the gamete is the sperm cell. In females, the gamete is the secondary oocyte.

When a sperm and oocyte fuse at fertilization, 23 plus 23 gives you 46, and we are back to diploid. That number — 46 — is the magic number for a human being. Anything more or less, and we have what is called a chromosomal aberration. We will come right back to that in a minute.

Slide 20: Mutations (0:28 – 0:30 • 2 min)

A mutation is any change in the DNA sequence. Mutations come in many flavors — some are tiny, some are enormous.

- Point mutations change a single base. Sometimes that change is silent — the protein comes out the same. Sometimes it changes one amino acid. And sometimes it creates a stop signal that cuts the protein short.
- Insertions and deletions add or remove bases. If the number added or removed is not a multiple of three, you get a frameshift, and everything downstream is read wrong. Those tend to be very destructive.
- Larger mutations can affect whole sections of a chromosome — duplications, deletions, inversions, and translocations.

And mutations come from somewhere. Some are spontaneous — random copying errors during DNA replication. Some are induced by mutagens — radiation, certain chemicals, viruses. Some are inherited from a parent. The clinical question is always: what was the source, where did the mutation land, and what protein is now broken?

Slide 21: Chromosome Aberrations (0:30 – 0:31 • 1 min)

When the change is at the level of an entire chromosome — not just a base or a stretch of bases — we call that a chromosomal aberration.

Two big buckets. First, structural aberrations, where the chromosome itself is rearranged. Pieces can be deleted, duplicated, flipped — that's an inversion — or swapped between chromosomes — that's a translocation.

Second, numerical aberrations — also called aneuploidy — where the number of chromosomes is wrong. Either there is an extra one, which is most common, or one is missing.

Slide 22: Examples of Aneuploidy (0:31 – 0:33 • 2 min)

Let me run through the classic examples — these are the ones you will see again and again on board exams and in clinical practice.

- Down syndrome — trisomy 21. An extra copy of chromosome 21. Autosomal, meaning it is one of the non-sex chromosomes. The most common chromosomal cause of intellectual disability and the one you are most likely to encounter clinically.

- Patau syndrome — trisomy 13. Severe, often incompatible with long-term survival.
- Edwards syndrome — trisomy 18, also severe (not on the slide but worth knowing).
- Trisomy X — three X chromosomes in a female. Often subtle, sometimes asymptomatic.
- Klinefelter syndrome — an XXY male. Extra X chromosome. Tall stature, often reduced fertility, sometimes learning differences.
- Turner syndrome — a single X chromosome in a female, written as 45,X. Short stature, often heart and kidney findings, infertility.
- Partial trisomy — only part of an extra chromosome is present, not the whole thing.
- Chromosomal mosaicism — some cells have the abnormal chromosome count and others do not, which often softens the clinical picture.

If you walk away with one thing from this slide, walk away with: Down is 21, Patau is 13, Edwards is 18, Turner is one X, Klinefelter is XXY. Those five facts get you through a lot of test questions.

Slide 23: Inheritance Terminology (0:33 – 0:34 • 1 min)

Quick vocabulary blitz. These are words that have to be second-nature.

- Gene — a stretch of DNA that codes for one product, usually one protein.
- Allele — one of the alternative versions of a gene. You inherit one allele from each parent.
- Recessive — only expressed when both alleles are the same recessive version.
- Dominant — expressed even if you only have one copy.
- Heterozygous — two different alleles. Homozygous — two of the same.
- Genotype — the genetic code itself. Phenotype — the observable trait.
- Penetrance — what percentage of people with the genotype actually show the phenotype.
- Expressivity — among the people who show it, how strongly do they show it.

Penetrance and expressivity tend to trip students up, so here is a one-line memory hook: penetrance is whether, expressivity is how much.

Slide 24: Modes of Inheritance (0:34 – 0:36 • 2 min)

Four classic modes of inheritance you should be able to recognize from a pedigree.

- Autosomal dominant — the trait shows up in every generation, both sexes equally affected. One copy of the bad allele is enough. Examples: Huntington disease, Marfan, neurofibromatosis.
- Autosomal recessive — often skips generations because two unaffected carriers can have an affected child. Both sexes equally affected. Examples: cystic fibrosis, sickle cell, Tay-Sachs.
- X-linked recessive — males much more often affected because they only have one X. The classic example: hemophilia, and Duchenne muscular dystrophy.

- X-linked dominant — rarer; both sexes affected, but no father-to-son transmission, because dad gives sons his Y.

On a pedigree, the easiest tell for X-linked anything: zero father-to-son transmission. If you see a man passing it to his son, it is not X-linked.

Slide 25: Understanding Epigenetics (0:36 – 0:38 • 2 min)

Now here is where it gets fun. Epigenetics.

The word breaks apart into “epi,” meaning above, and “genetics.” So epigenetics is, literally, the layer of regulation that sits above the DNA sequence. The DNA letters do not change — but the marks on top of the DNA, and the way it is folded and packaged, change which genes get read and which stay quiet.

There are three main mechanisms to know.

- DNA methylation — small chemical tags placed on the DNA, usually silencing genes.
- Histone modification — the proteins that DNA wraps around can be marked, either tightening or loosening the DNA so it is harder or easier to read.
- Non-coding RNA — small RNA molecules that block specific mRNAs from being translated.

Why does this matter? Because epigenetic marks change in response to your environment. Diet, stress, smoking, exposures during pregnancy — all of these leave marks. And those marks change which proteins your cells make. That is why two identical twins, with the same DNA, can have different health outcomes — they live different lives, and their epigenomes diverge over time.

This is also why some diseases — type 2 diabetes, certain cancers, some psychiatric conditions — cluster in families even when no clean Mendelian gene is at work. Genetics loaded the gun. Epigenetics, environment, and behavior pull the trigger.

Slide 26: Genetic Disorders — Wrap-up (0:38 – 0:39 • 1 min)

So to recap this section: genetic disorders can come from a single base change, a chromosomal aberration, an inheritance pattern, or epigenetic regulation. Sometimes more than one of those is in play in the same patient.

As a nurse, your job is not to diagnose the precise mutation — that is the geneticist’s job. Your job is to recognize the pattern, ask good family-history questions, advocate for genetic counseling when appropriate, and educate your patient about what the diagnosis means for them and their family.

Section 4 • Knowledge Checks — Genetics (0:39 – 0:43)

Slide 27: Knowledge Check — Down Syndrome & Maternal Age (0:39 – 0:41 • 2 min)

Time for our first knowledge check. Let me read it.

A 35-year-old primigravida — meaning, pregnant for the first time — asks about her risk of having a child with Down syndrome compared with a 25-year-old. The best explanation is — A, B, C, or D? Drop your answer in the chat.

[Stage cue] Read each option aloud. Then pause 20–30 seconds. Read responses from chat.

Alright — the correct answer is B. Aging oocytes spent longer arrested in meiosis, and their spindle apparatus is more error prone.

Here is the why. Female oocytes begin meiosis during fetal development, and then they pause — they arrest in prophase I — until that oocyte is ovulated, decades later. The longer that pause, the more wear and tear the meiotic spindle accumulates. When meiosis finally resumes, the spindle is more likely to mis-segregate a chromosome — that is called nondisjunction — and you end up with an oocyte that has 24 chromosomes instead of 23. Fertilize that, and you get trisomy.

Why are the others wrong? A is about sperm, but maternal age — not paternal age — is the major driver here. C, environmental DNA damage during pregnancy, is not the primary mechanism. And D is wrong because Down syndrome is a trisomy, not a recessive inheritance.

Slide 28: Knowledge Check — Twins & Type 2 Diabetes (0:41 – 0:43 • 2 min)

Knowledge check number two.

Identical twins have the same DNA. One develops type 2 diabetes at age 45. The other does not. Which explanation best accounts for this difference? A through D — go ahead and drop your answer.

[Stage cue] Read the options aloud. Pause for 20–30 seconds.

Correct answer is B — different environmental exposures produced different epigenetic marks that altered gene expression.

Same DNA, different lives equals epigenetic divergence. Diet, exercise, sleep, stress, smoking — all of those leave methylation patterns and histone modifications that change which genes are switched on and off. One twin's pancreas might end up expressing genes that protect insulin-producing cells; the other twin's might not.

Why are the others wrong? A is wrong because the actual DNA sequence does not change — that would be a mutation. C is wrong because identical twins start with the same DNA — they cannot have a

different gene from each other. And D contradicts the question — the question tells you they are identical.

Two for two on the section. Beautiful. Let's keep going.

Section 5 • Cellular Adaptation & Injury (0:43 – 0:50)

Slide 29: Section Objective — Altered Cell & Tissue Biology (0:43 – 0:44 • 1 min)

Section objective for the next chunk: define altered states of both cellular and tissue biology.

This is where pathophysiology becomes very visible. The names we are about to learn — atrophy, hypertrophy, hyperplasia, dysplasia, metaplasia — are words you will see all over your patients' charts. Pathology reports, imaging reports, biopsy results — they are full of these words. So this is your chance to make them stick.

Slide 30: Cellular Adaptation (0:44 – 0:47 • 3 min)

Cells are not passive. When they are stressed, they adapt. There are five classic adaptive responses, and you should be able to define every single one of them.

- Atrophy — cells get smaller. Decreased size. Think of a leg that has been in a cast for six weeks. The muscle was unused, and the fibers shrank. The cells are still alive, just smaller.
- Hypertrophy — cells get bigger. Increased size, no extra cells. Think of the heart in long-standing hypertension — the cardiac muscle cells, which cannot divide, simply grow larger to handle the increased pressure.
- Hyperplasia — more cells. Increased number, not size. Think of a callus on your hand from gardening — your skin makes more cells in response to friction. Or think of the prostate enlarging with age.
- Dysplasia — disorganized cells. Cells that look abnormal in shape and arrangement, often a precursor to cancer. Think of the cervical dysplasia we screen for with a Pap smear.
- Metaplasia — one cell type replaced by another. The replacement cells are usually better suited to the stress, even if they belong somewhere else. Think of a long-time smoker's airway — the delicate ciliated columnar cells are replaced by tougher squamous cells.

A trick to keep them straight. A and H — atrophy and hypertrophy — change size. Hyperplasia changes number. Dysplasia changes shape. Metaplasia changes type.

Atrophy and hypertrophy and hyperplasia are reversible if the stress goes away. Dysplasia can be reversible early on but is worrisome because it can progress to cancer. Metaplasia is reversible too — but if the stress continues, metaplastic tissue can also progress to dysplasia and then cancer. Smoke long enough and squamous metaplasia in the airway becomes squamous cell carcinoma.

Slide 31: Cellular Injury and Death (0:47 – 0:50 • 3 min)

If the stress overwhelms the cell's ability to adapt, the cell becomes injured. And if the injury is severe enough, the cell dies.

- Hypoxia — low oxygen at the cell. The single most common cause of cellular injury you will encounter clinically. Without oxygen, mitochondria cannot make ATP, the sodium-potassium pump fails, the cell swells, calcium floods in, and a destructive cascade begins.
- Necrosis — uncontrolled, traumatic cell death. The cell ruptures, releases its contents into the surrounding tissue, and triggers inflammation. Necrosis is messy. Think of a heart attack — the dead myocardium is a necrotic event.
- Apoptosis — programmed, controlled cell death. The cell shrinks, packages itself up neatly, and gets cleaned up by neighbors and immune cells with no inflammation. Apoptosis is quiet. It is how we get rid of cells we no longer need — and it is how the body removes cells with damaged DNA before they become cancer.

So here is the story arc for any cell. Stress arrives. The cell adapts — atrophy, hypertrophy, hyperplasia, dysplasia, metaplasia. If the stress continues, the adaptation fails, and the cell becomes injured — most often through hypoxia. If injury is mild, the cell can recover. If injury is severe, the cell dies — quietly, by apoptosis, or violently, by necrosis.

Hold onto that arc. It is the through-line for every disease process you will study this semester.

Section 6 • Fluids, Electrolytes, Acid-Base (0:50 – 0:57)

Slide 32: Fluid Balance and Edema (0:50 – 0:52 • 2 min)

Now to fluids. About 60 percent of your body weight is water. That water lives in two main compartments — inside cells, called intracellular, and outside cells, called extracellular. The extracellular space is further split into the fluid between cells, called interstitial, and the fluid in your blood vessels, called intravascular.

Water moves between these compartments along pressure gradients. Two pressures matter most. Hydrostatic pressure — think of it as water trying to push out of the blood vessel. And oncotic pressure — driven mostly by albumin — think of it as water trying to be pulled back in.

Edema — swelling — happens when that balance fails. Four classic mechanisms.

- Increased hydrostatic pressure — heart failure, where blood backs up in the veins.
- Decreased oncotic pressure — low albumin, as in liver failure or severe malnutrition.
- Increased capillary permeability — inflammation.
- Lymphatic obstruction — surgical lymph node removal, infection.

When you see swelling on a patient, run through those four. One of them, or some combination, is the answer.

Slide 33: Alterations in Fluid Balance — Electrolytes (0:52 – 0:54 • 2 min)

Three electrolytes you must know cold for nursing practice — sodium, potassium, calcium.

- Sodium — the major extracellular cation. It largely determines plasma osmolality and fluid distribution. Hyponatremia, low sodium, often presents with confusion, headache, and in severe cases, seizures. Hypernatremia, high sodium, presents with thirst, restlessness, and in severe cases, neurologic decline.
- Potassium — the major intracellular cation. The big one for the heart. Hypokalemia, low potassium, gives you weakness, cramps, and dangerous cardiac arrhythmias — flattened T waves and U waves on the ECG. Hyperkalemia, high potassium, gives you peaked T waves and can stop the heart. Both extremes are emergencies.
- Calcium — bone health, neuromuscular function, clotting. Hypocalcemia gives you tingling around the mouth, muscle cramps, and a positive Trousseau or Chvostek sign. Hypercalcemia gives you the classic “stones, bones, groans, and psychiatric overtones” — kidney stones, bone pain, abdominal pain, and altered mental status.

If I had to give you one clinical heuristic for tonight, it would be this: any time your patient has new confusion, new weakness, or new arrhythmia, the first thing to check is their electrolytes.

Slide 34: Acid-Base Balance (0:54 – 0:55 • 1 min)

Now we shift into acid-base. The body is obsessed with pH. The normal arterial pH is 7.35 to 7.45 — a very narrow window. Below 7.35 is acidemia. Above 7.45 is alkalemia. Outside the range of about 6.8 to 7.8, you cannot survive.

Two systems regulate pH. The lungs handle CO_2 — fast, within minutes. Breathe faster, blow off more CO_2 , raise pH. The kidneys handle bicarbonate — slow, over hours to days. Reabsorb more bicarb, raise pH.

CO_2 is the acid side. Bicarb is the base side. Keep that straight and acid-base questions become much simpler.

Slide 35: Acid-Base Imbalances (0:55 – 0:56 • 1 min)

Four primary disorders. Memorize the table.

- Respiratory acidosis — pH low, CO_2 high. The lungs are not blowing off enough CO_2 . Causes: COPD, opioid overdose, neuromuscular weakness.
- Respiratory alkalosis — pH high, CO_2 low. The patient is breathing off too much CO_2 . Causes: anxiety/hyperventilation, early salicylate overdose, pulmonary embolism.
- Metabolic acidosis — pH low, bicarb low. Causes: diabetic ketoacidosis, lactic acidosis, severe diarrhea, kidney failure.
- Metabolic alkalosis — pH high, bicarb high. Causes: vomiting, excess antacids, certain diuretics.

Each primary disorder gets a compensatory response from the other system. Lungs are fast — minutes. Kidneys are slow — days.

Slide 36: ROME Mnemonic (0:56 – 0:57 • 1 min)

And here is the mnemonic that will save you on every acid-base question for the rest of your life. ROME.

- R-O — Respiratory Opposite. In respiratory disorders, the pH and the CO_2 move in opposite directions.
- M-E — Metabolic Equal. In metabolic disorders, the pH and the bicarb move in the same direction.

So look at the lab. Is pH and CO_2 moving opposite — respiratory. Is pH and bicarb moving together — metabolic. Then look at whether pH is high or low to label acidosis or alkalosis.

There is a great short video linked on this slide that walks through ROME with examples. I would highly recommend watching it before the quiz. The link is in the notes — about 5 minutes.

Section 7 • Final Knowledge Checks & Wrap-Up (0:57 – 1:00)

Slide 37: Knowledge Check — Left Ventricular Wall Thickening (0:57 – 0:58 • 1 min)

Knowledge check number three.

A patient with long-standing hypertension has thickening of the left ventricular wall on echocardiogram. This change represents — A, hyperplasia; B, hypertrophy; C, metaplasia; D, atrophy. Drop your answer in the chat.

[Stage cue] Pause 15–20 seconds.

Correct answer is B — hypertrophy. Cardiac myocytes are post-mitotic, which means they cannot divide. So when chronic pressure load arrives — like in long-standing hypertension — the cells respond the only way they can: by getting bigger. That is hypertrophy.

A, hyperplasia, would require the cells to divide and add more cells, which cardiac muscle cells cannot do. C, metaplasia, would require switching cell types — also no. D, atrophy, is shrinkage — the opposite of what we see here.

Slide 38: Knowledge Check — DKA Acid-Base (0:58 – 0:59 • 1 min)

Last knowledge check of the night.

A patient with DKA — diabetic ketoacidosis — has a pH of 7.22, a PaCO₂ of 26, and a bicarbonate of 10. The acid-base disturbance is — A, B, C, or D?

[Stage cue] Pause 20–30 seconds.

Correct answer is C — metabolic acidosis with respiratory compensation.

Walk through it with me. Step one: pH is 7.22 — that is low — so this is an acidosis. Step two: bicarbonate is 10 — also low. Bicarb is moving the same direction as pH — that is the M-E half of ROME, metabolic equal. So it is a metabolic acidosis. Step three: PaCO₂ is 26 — that is low. The lungs are blowing off CO₂ as fast as they can, trying to drag the pH back up. That is respiratory compensation, and clinically you would see this as Kussmaul breathing — those deep, fast, hungry breaths.

Beautiful. That is exactly the kind of thinking I want from you on the quiz.

Wrap-Up & Closing (0:59 – 1:00)

[No specific slide — close on the final knowledge check or return to the title slide.]

Alright — we are at the end of our hour. Let's recap, very quickly.

- Pathophysiology is the study of how the body's functions change in disease — and that change always happens to a human being who is suffering.
- Genes, mutations, chromosomal aberrations, and inheritance patterns set the stage; epigenetics is the dimmer switch that turns genes up and down based on environment and behavior.
- Cells adapt — atrophy, hypertrophy, hyperplasia, dysplasia, metaplasia — and when adaptation fails, they get injured, often by hypoxia, and they die — quietly by apoptosis or violently by necrosis.
- Fluid balance, sodium, potassium, calcium, and acid-base — these are the daily tools you will use at the bedside. Memorize ROME. It will save your shift.

Before you log off — three things. One, take the Week 1 quiz before Sunday at 11:59 p.m. Mountain. Two, finish the case study. Three, email me if anything in the readings does not click — I would much rather hear from you in week one than in week six.

If you have a quick question right now, drop it in the chat and I will stick around for the next few minutes. Otherwise — thank you for being here, thank you for being engaged, and I will see you next week.

[Stage cue] *Stop recording when the room is empty. Save the chat transcript if your platform supports it.*